N=32 FP 32 (4 bytes)

1) Naive triple loop:

In B each element is accessed 32 times (N).

A: M×N×K ops :- 32×32×32 = 32,768 floats

B: 32×32×32 = 32,768 floats.

Total element accessed = 65,536 access

Total DRAM traffic :-

65,536 × 4 bytes = 262,144 bytes (256 KB)

2) Tiled loop :- T = 8

A = M×N = 32×32 = 1024 float

B. 32×32 = 1024 float

Total element accessed = 2048

Total DRAM traffic :-

2048×4 = 8192 bytes

c) Ratio = 262144/8192 = 32

The ratio equals N because the Naive approach reads every element from DRAM N times, whereas the tiled approach reads exactly once.

d) Compute : 10 TFLOPS

Bandwidth : 320 GB/s

$AI = \frac{10{,}000 \times 10^9}{320 \times 10^9} = 31.25$ FLOP's / byte

Total FLOP's = 2×32^3 = 65,536 FLOP's

1) Naive execution time

Traffic 262,144 bytes

AI : 65,536 / 262,144 = 0.25 FLOP's byte

Because 0.25 << 31.25 it is memory bound

Execution time: Bec it is memory bound time is dictated by bandwidth

262,144 bytes / 320 GB/s = 819.2 ns

2) Tiled execution time.

Traffic :- 8,192 bytes

A.I = 65,536 / 8192 = 8 FLOP / byte

Because 8 < 31.25, the tiled approach is memory bound even though it is much closer to ridge point

Execution time :- 8,192 bytes / 320 GB/s

= 25.6 ns